UNIT 5 · 8 hours

Lag Compensator



Compensators, Controllers and State-Space Analysis

Control Systems (22EEG53)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

U5

Learning objectives

- Write the transfer function and Bode plot of a lag compensator
- Explain its effect on steady-state error, bandwidth and margins
- Design a lag compensator using the Bode plot
- Verify the design

Syllabus context

Compensators and controllers: introduction to compensators, lag, lead and lag-lead compensators, P, PI and PID controllers.

U5

Lag compensator

$$G_c(s) = (1 + Ts)/(1 + \beta Ts),\ \beta > 1$$

zero at −1/T, pole at −1/(βT)

$$\varphi_m = -\sin^{-1}[(\beta - 1)/(\beta + 1)]$$

max phase lag

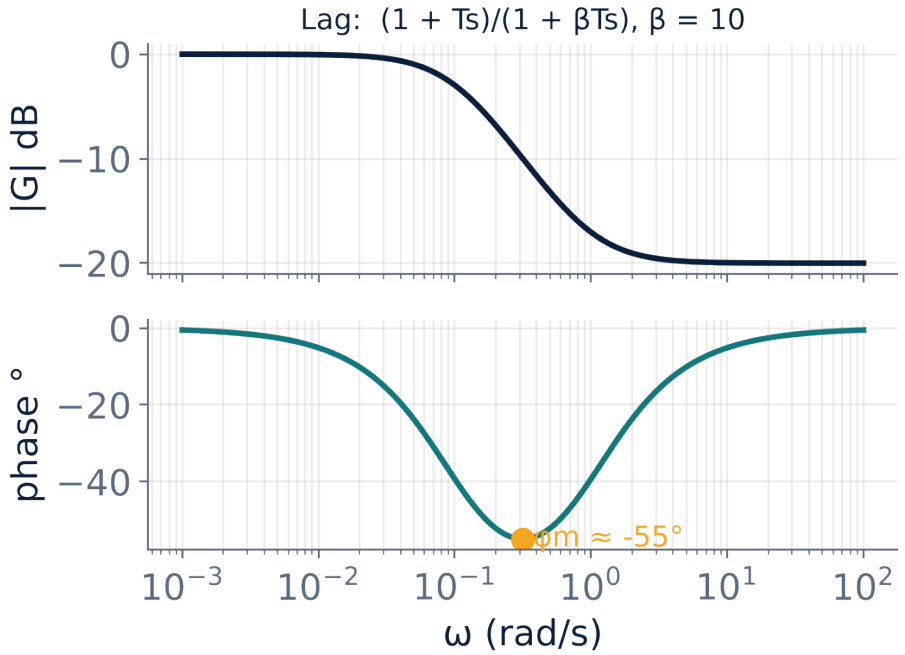
$$\omega_m = 1/(T\sqrt{\beta})$$

frequency of max lag

- Pole is closer to the origin than the zero.
- DC gain = 1; high-frequency gain = 1/β: attenuation 20 log β dB.

U5

Bode plot of a lag network



- Gain falls by 20 log β dB between 1/(βT) and 1/T.
- Phase negative with a minimum of −55° for β = 10.
- Place the corners well below the gain crossover so the phase lag there is small.

U5

Effect of lag compensation

Steady-state accuracy

Allows higher low-frequency gain (Kv up by up to β) without changing high-frequency behaviour.

Stability / transient

Reduces gain at crossover to restore PM; lowers bandwidth so response becomes slower.

Cautions

Noise attenuation improves; the extra pole near the origin gives a slow tail in the step response.

U5

Bode design procedure (lag)

1

Set K

Choose K to meet the steady-state error (Kv).

2

Find new crossover

Find ω where phase = −180° + PM + 5°–12°.

3

Choose β

β = |G(jω)| at that frequency (attenuation needed).

4

Choose T

Zero at 1/T = ωnew/10 (a decade below).

5

Check

Verify PM, GM and steady-state error.

Compensator: $G_c(s) = (1 + Ts)/(1 + \beta Ts)$ with $\beta = |G(j\omega_{new})|$ and $1/T = \omega_{new}/10$.

U5

Numerical: lag design

Problem G(s) = K/[s(1 + s)(1 + 0.5s)], Kv = 5 required, PM ≥ 40°. Design a lag compensator.

- K = Kv = 5. Uncompensated: ωgc ≈ 1.8 rad/s, PM ≈ −13° (unstable)
- Need phase ≈ −180° + 40° + 5° = −135° at the new crossover ⇒ ω ≈ 0.56 rad/s
- |G(j0.56)| = 7.5 ⇒ β = 7.5; 1/T = 0.056 ⇒ T = 17.9 s
- Gc = (1 + 17.9s)/(1 + 134s); phase of Gc at 0.56 ≈ −4.9°

Answer: Compensated PM ≈ 40°, Kv = 5 retained

U5

Key takeaways

- Lag: $G_c = (1 + Ts)/(1 + \beta Ts)$, $\beta > 1$
- Improves steady-state accuracy; reduces bandwidth
- Place the corner a decade below the new gain crossover
- β = attenuation required at the new crossover

U5

Practice questions

Q1. Draw the pole-zero plot and Bode plot of a lag compensator.

Q2. State the effects of a lag compensator on time response.

Q3. Find φm and ωm for T = 1, β = 10.

Q4. Outline the Bode design procedure for a lag compensator.